EV Adoption by Electric Utility: Footprint Metrics and Recent-Adoption Patterns (Model-Year Cohorts)

Based on the Washington Electric Vehicle Population dataset

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1. Project Overview

This project analyzes electric vehicle adoption using Electric Utility as the main grouping variable. The goal is to:

1. compute a utility-level EV footprint that summarizes adoption patterns, including EV count, BEV/PHEV composition, and range statistics;
2. identify utility territories with the highest recent-adoption share using model-year cohorts as a proxy for adoption recency because direct registration-date fields are not consistently available in the dataset.

This project turns row-level EV records into utility-level summaries so different electric utility territories can be compared more clearly.

2. Dataset Description

I used the Washington Electric Vehicle Population Data file with 276,828 rows. The analysis relies on these key fields:

1. Electric Utility as the grouping key
2. Electric Vehicle Type to distinguish BEV and PHEV records
3. Electric Range for capability-related summary statistics
4. Model Year for recency and cohort analysis
5. County, City, and Postal Code for coverage context

Basic dataset checks:

1. Total rows: 276,828
2. Missing Electric Utility in raw data: 15
3. Standardized utility categories: 78
4. Missing Model Year: 0
5. Model Year range: 1999 to 2026
6. Electric Range = 0 share: 63.4%

3. Data Cleaning and Feature Engineering

The script uses a simple cleaning pipeline:

1. It keeps only the columns needed for the analysis.
2. It cleans text fields by uppercasing, trimming spaces, and standardizing extra whitespace.
3. It standardizes Electric Utility values and labels missing values as `UNKNOWN UTILITY`.
4. It converts Model Year, Electric Range, and Postal Code to numeric values.
5. It creates `Is BEV`, `Is PHEV`, and `Is Recent Model Year` as helper columns.
6. It creates `Range Positive` so range statistics can also be calculated without zero values.
7. It groups model years into five cohort bins: 2014 and earlier, 2015-2017, 2018-2020, 2021-2023, and 2024+.

The following function standardizes utility names before grouping. Its job is to make sure small formatting differences do not split the same utility into multiple categories.

```
def clean_utility_name(value):
    if pd.isna(value):
        return "UNKNOWN UTILITY"

    text = str(value).upper().strip()
    text = text.replace("||", "|")

    pieces = []
    for part in text.split("|"):
        part = " ".join(part.split())
        if part:
            pieces.append(part)

    if not pieces:
        return "UNKNOWN UTILITY"
    return " | ".join(pieces)
```

The next function handles most of the cleaning work. It selects the needed columns, cleans text fields, converts numeric columns, and creates the helper columns used later in the analysis.

```
def clean_data(raw_df, recent_year):
    df = raw_df[REQUIRED_COLUMNS].copy()

    text_cols = [
        "County",
        "City",
        "State",
        "Make",
        "Model",
        "Electric Vehicle Type",
        "Clean Alternative Fuel Vehicle (CAFV) Eligibility",
    ]
```

```
    for col in text_cols:
        df[col] = df[col].fillna("UNKNOWN").astype(str).str.upper().str.strip()
        df[col] = df[col].str.replace(r"\s+", " ", regex=True)

    df["Utility Standardized"] = df["Electric Utility"].apply(clean_utility_name)
    df["Model Year"] = pd.to_numeric(df["Model Year"], errors="coerce").astype("Int64")
    df["Electric Range"] = pd.to_numeric(df["Electric Range"], errors="coerce")
    df["Postal Code"] = pd.to_numeric(df["Postal Code"], errors="coerce")

    df["Is BEV"] = df["Electric Vehicle Type"] == "BATTERY ELECTRIC VEHICLE (BEV)"
    df["Is PHEV"] = df["Electric Vehicle Type"] == "PLUG-IN HYBRID ELECTRIC VEHICLE (PHEV)"
    df["Is Recent Model Year"] = df["Model Year"] >= recent_year

    # Keep positive range separately because a lot of rows use 0.
    df["Range Positive"] = df["Electric Range"].where(df["Electric Range"] > 0)

    df["Model Year Cohort"] = pd.cut(
        df["Model Year"].astype(float),
        bins=[0, 2014, 2017, 2020, 2023, np.inf],
        labels=COHORT_LABELS,
    )

    return df
```

4. Methodology

4.1 Utility Footprint Metrics

Using utility-level grouped summaries, the analysis computes:

1. total EV count
2. BEV and PHEV counts and shares
3. average and median electric range
4. average and median positive electric range
5. minimum, median, and maximum model year
6. county count and city count

This aggregation function creates the main utility-level summary table. It is the part of the script that turns row-level vehicle records into one summary row per standardized utility.

```
def make_utility_summary(clean_df):
    grouped = clean_df.groupby("Utility Standardized", dropna=False)

    summary = grouped.agg(
        total_ev_count=("DOL Vehicle ID", "count"),
        bev_count=("Is BEV", "sum"),
```

```
        phev_count=("Is PHEV", "sum"),
        recent_model_year_count=("Is Recent Model Year", "sum"),
        average_electric_range=("Electric Range", "mean"),
        median_electric_range=("Electric Range", "median"),
        average_positive_electric_range=("Range Positive", "mean"),
        median_positive_electric_range=("Range Positive", "median"),
        min_model_year=("Model Year", "min"),
        median_model_year=("Model Year", "median"),
        max_model_year=("Model Year", "max"),
        county_count=("County", "nunique"),
        city_count=("City", "nunique"),
    ).reset_index()

    summary["bev_share"] = summary["bev_count"] / summary["total_ev_count"]
    summary["phev_share"] = summary["phev_count"] / summary["total_ev_count"]
    summary["recent_model_year_share"] = (
        summary["recent_model_year_count"] / summary["total_ev_count"]
    )

    column_order = [
        "Utility Standardized",
        "total_ev_count",
        "bev_count",
        "bev_share",
        "phev_count",
        "phev_share",
        "recent_model_year_count",
        "recent_model_year_share",
        "average_electric_range",
        "median_electric_range",
        "average_positive_electric_range",
        "median_positive_electric_range",
        "min_model_year",
        "median_model_year",
        "max_model_year",
        "county_count",
        "city_count",
    ]

    summary = summary[column_order]
    summary = summary.sort_values(["total_ev_count", "bev_share"], ascending=[False, False])
    return summary
```

4.2 Utilities With Highest Recent-Adoption Share

To estimate how recent each utility's EV fleet appears, the report uses:

`recent_model_year_share = share of vehicles with Model Year >= 2022`

Only utilities with at least 100 EVs are included in this ranking so that very small groups do not dominate the comparison.

This ranking function filters out very small utility groups and then sorts the remaining utilities by recent-model share. That is how the report identifies which utility territories appear most weighted toward newer EV model years.

```
def make_recent_adoption_ranking(utility_summary, min_utility_count):
    ranking = utility_summary[utility_summary["total_ev_count"] >=
min_utility_count].copy()
    ranking["recent_adoption_score"] = ranking["recent_model_year_share"]
    ranking = ranking.sort_values(
        ["recent_adoption_score", "recent_model_year_count", "total_ev_count"],
        ascending=[False, False, False],
    ).reset_index(drop=True)
    ranking.insert(0, "recent_adoption_rank", range(1, len(ranking) + 1))
    return ranking
```

5. Results

5.1 Overall Summary

Across the full dataset, the BEV share is 80.0%, which shows that battery-electric vehicles dominate the observed EV population. The statewide recent-model share is 69.3%, meaning a large portion of the fleet falls in model year 2022 or newer. The largest utility territory is PUGET SOUND ENERGY INC | CITY OF TACOMA - (WA) with 98,542 EVs, a BEV share of 82.6%, and a recent-model share of 73.2%.

5.2 Largest Utility Territories (Top 5 by EV Count)

Largest Utility Territories

Utility	EV Count	BEV Share	Recent Share
PUGET SOUND ENERGY INC	CITY OF TACOMA - (WA)	98,542	82.6%	73.2%
PUGET SOUND ENERGY INC	58,233	81.2%	69.4%
CITY OF SEATTLE - (WA)	CITY OF TACOMA - (WA)	45,996	80.3%	67.1%
BONNEVILLE POWER ADMINISTRATION	PUD NO 1 OF	16,614	76.0%	68.2%

CLARK COUNTY - (WA)			
BONNEVILLE POWER ADMINISTRATION | CITY OF TACOMA - (WA) | PENINSULA LIGHT COMPANY	12,590	77.3%	66.2%

Interpretation: the largest utility territories are concentrated among a small number of major utility combinations, and the top territories also tend to show relatively high BEV shares and relatively recent model-year composition.

5.3 Utility Territories With Highest Recent-Adoption Share (Top 5 by Recent-Model Share, min count = 100)

Utilities With Highest Recent-Adoption Share

Utility	EV Count	Recent Share	BEV Share
BONNEVILLE POWER ADMINISTRATION | CITY OF CENTRALIA - (WA) | CITY OF TACOMA - (WA)	315	73.3%	39.4%
CITY OF TACOMA - (WA) | TANNER ELECTRIC COOP	388	73.2%	86.3%
PUGET SOUND ENERGY INC | CITY OF TACOMA - (WA)	98,542	73.2%	82.6%
BONNEVILLE POWER ADMINISTRATION | VERA IRRIGATION DISTRICT #15	615	72.8%	56.4%
BONNEVILLE POWER ADMINISTRATION | AVISTA CORP | INLAND POWER & LIGHT COMPANY	4,727	70.2%	72.9%

Interpretation: utility size and recency are related but not identical. Some very large utilities also rank highly by recent-model share, while other smaller territories appear especially recent because a large fraction of their fleets are concentrated in newer cohorts.

7. Limitation

This analysis has several important limitations:

1. Model year is not registration year, so recent-model share is only a proxy for recency and not a direct growth rate.
2. Electric Range contains many zeros; 63.4% of records have `Electric Range = 0`, which can distort raw range averages.
3. Some Electric Utility values contain multiple utility names in a single field, which preserves the source encoding but can blur territory interpretation.
4. The analysis is descriptive and does not test causal drivers such as charging infrastructure, policy, or demographic differences.

8. Reproducibility and Outputs

This project is reproducible because the analysis is implemented in a single Python script that reads the raw CSV, creates the derived columns, calculates grouped summaries, and writes labeled output files. The script produces:

1. `cleaned_ev_population.csv`
2. `data_quality_summary.csv`
3. `utility_footprint_summary.csv`
4. `recent_adoption_utilities.csv`
5. `model_year_cohort_distribution.csv`
6. `top_utilities_by_ev_count.svg`
7. `top_utilities_by_recent_adoption_share.svg`

9. Conclusion

Overall, this project transforms row-level EV registration data into a utility-level analytical view that is easier to compare across territories. By combining EV footprint metrics with a cohort-based recency ranking, the report highlights both where EVs are most concentrated and which utility territories appear most weighted toward newer vehicles. Future work could extend the analysis with registration-date data, external infrastructure variables, or dashboard-based visualization.